

Determination of the Angular Distribution of Muons as Secondary Cosmic Rays

Adam Quayle

Department of Physics, Maynooth University

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ABSTRACT

The angular distribution of secondary cosmic muons was determined to be almost isotropic with slight attenuation at larger zenithal angles due to atmospheric scattering and absorption. It was also determined that approximating the solid angle subtended by a rectangular plate using a spherical cap is not accurate enough to successfully determine this angular distribution.

Introduction

Primary cosmic rays are high-energy ($> 10^6$ eV) particles of extra-solar origin which bombard the Earth's atmosphere from all directions. Upon colliding with the atmosphere, the kinetic energy of the incident particle is converted into the mass of new particles, which cascade down, eventually reaching the ground. The particles created in this avalanche, secondary cosmic rays, can be detected by suitable apparatus at the surface.

The majority of primary cosmic rays are completely ionised nuclei which are ejected during high-energy events in the Universe. Due to the charge of these particles, their trajectories are bent unpredictably by any diffuse magnetic fields on their path to Earth, and so their angular distribution should be isotropic. A slight anisotropy, however, has been noted in those cosmic rays of extraordinary energy, which are now theorised to be of extragalactic origin (Pierre Auger Collaboration, 2017).

The purpose of the experiment in this work is to investigate the angular distribution of muons in the secondary cosmic ray population. The particle shower created by an incident primary cosmic ray can spread up to 100 m across by the time it reaches the surface. Due to this relative localisation, secondary cosmic rays detected at the surface can give useful information about the angular distribution of *all* cosmic rays, not only those which are considered secondary.

Method

In this experiment, plastic scintillators are used as the initial detectors for muon events. Plastic scintillators are advantageous for the purpose of detecting cosmic rays, as they produce a large amount of light in a very short duration pulse during excitation. (Sekiya *et al.*, 2022). This allows many events to be detected in quick succession without blending together, as is required when detecting muons in a high-density particle cascade.

Two such plastic scintillators are placed above each other with a gap between them. When a muon strikes one of the scintillator blocks, it releases a flash of light which is then picked up by a photomultiplier tube (PMT) facing into the block. If a muon strikes either of the scintillators, the attached PMT releases a pulse of current which then passes through a discriminator and amplifier. Firstly, the discriminator circuit filters out any signals which are deemed too weak to be trustworthy, then the verified signals are amplified for further analysis.

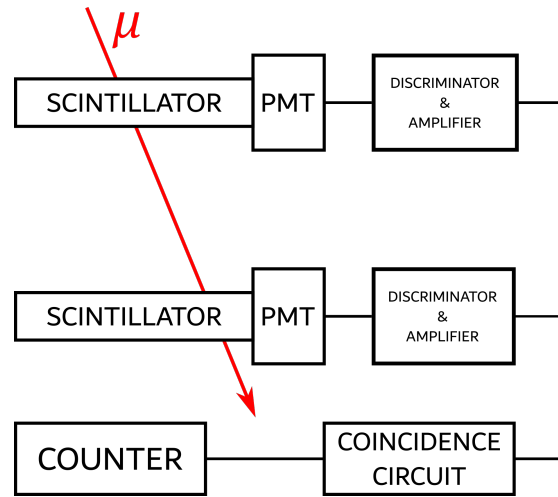


Figure 1: Schematic of the experiment

Due to the high flux of muons passing through the scintillator detectors, it is possible for two distinct muons to hit one scintillator each. In order to avoid this scenario causing a miscount, both detector branches are fed into a coincidence circuit. This circuit will test whether the two logic pulses from the amplifier are sufficiently coincident, and will discard the signals if it's likely that they came from two separate muons. This will ensure that all detected cosmic rays travel through *both* scintillators, and allow the muon detector to survey a specific part of the sky without stray muons interfering with the particle count.

Ultimately, if two events are detected within a short enough time, and the signal generated by their interaction with both scintillators is great enough, then the counter receives a signal from the detector and registers a count. Due to the above description of the coincidence circuit, only muons originating from the solid angle subtended by lines connecting the inner vertices of the two scintillators will be detected (as seen in Figure 2).

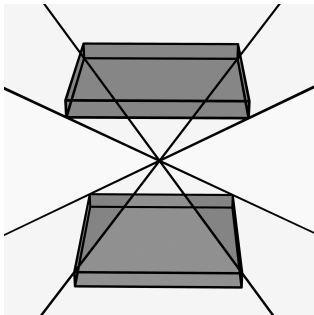


Figure 2: Maximum solid angle of muon detection.

It is possible for a muon to travel vertically through both scintillators and be detected, despite leaving the solid angle indicated in Figure 2. However, this is not an issue, as the detection of these muons is independent of the distance between the scintillators. These muons fall into a special case which contributes to a persistent background activity which doesn't change as the scintillators are separated.

The solid angle subtended by the scintillators can be approximated by constructing a spherical cap whose circular base is inscribed within the plane of the scintillator. This construction will cause a slight overestimation on the rate of incident muons, as it omits the fact that the edges of the scintillator can still detect particles. The scintillator plates are actually rectangular in reality, which also contributes to the error of this approximation.

The solid angle Ω subtended by the spherical cap described is given by

$$\Omega(d) = 2\pi \left(1 - \frac{d}{\sqrt{d^2 + L^2}} \right) \quad (1)$$

where d is the distance between the scintillator blocks and L is the side-length of a scintillator block.

Experimental Procedure

In the preparation of the experiment, confirming the working order of each component is crucial. If this is not done, problems with the apparatus during the experiment become very difficult to diagnose.

The Amplifier/Discriminator Circuits

The amplifier/discriminator circuits are checked with the aide of a fast-pulse module (FPM). In order to ensure correct operating order of this module, it is connected to an oscilloscope and put under load. An analogue pulse of about 18 ns will confirm that the battery-powered FPM is functioning as intended. Each amplifier/discriminator circuit is connected in series between the fast pulse module and the oscilloscope one at a time. The FPM is then supplied with power, and longer logic pulses of about $5.5 \mu\text{s}$ should appear on the oscilloscope, each representing a successful amplification of a sufficient pulse supplied by the FPM; this indicates that the amplifier/discriminator circuit is operative.

The Photomultiplier Tubes (PMTs)

The PMTs are supplied with a small voltage which is slowly increased until reaching the operational tension of 1.5 kV. During this process of charging the photocathodes within the PMTs, a microammeter is used to monitor the output current of the detector. If the current rises from zero unexpectedly, this indicates that light is leaking into the detector in some manner, causing the PMT to fire erroneously. When both PMTs are at full tension with no output current, each is connected to the oscilloscope. Short-duration spikes of current at random intervals should appear on the oscilloscope, each spike corresponds to a muon event within the scintillator.

The Coincidence Circuit

Once the PMTs are confirmed to be in working order, they are both connected to the amplifier/discriminator circuits. These circuits are then fed into a coincidence circuit which tests if the signal from each PMT overlaps sufficiently as to be considered the same detection.

To confirm the correct operation of the coincidence circuit, a long delay cable can be inserted between one of the amplifier/discriminator circuits and the coincidence circuit. This causes a temporal separation between the signals delivered for a single detection, meaning the coincidence circuit should discard the signals due to insufficient overlap. Observationally, this appears as a significant decrease in the number of detections appearing on the oscilloscope. The only detections that appear now are coincidental, consisting of two muons impacting the scintillators with the correct time delay to counteract the delay cable.

Conducting the Experiment

Once all components are confirmed to be in working order, the oscilloscope is disconnected and replaced with a digital counter. The distance between the scintillators is recorded for the purposes of calculating the solid angle subtended (Equation 1). The counter is then allowed to reach a pre-decided tally N (in this experiment, $N = 400$). The time Δt taken to reach this tally is recorded with a stopwatch, the mean muon count rate R for a solid angle Ω is then calculated via

$$R = \frac{N}{\Delta t} \quad (2)$$

Results

Preliminary Results

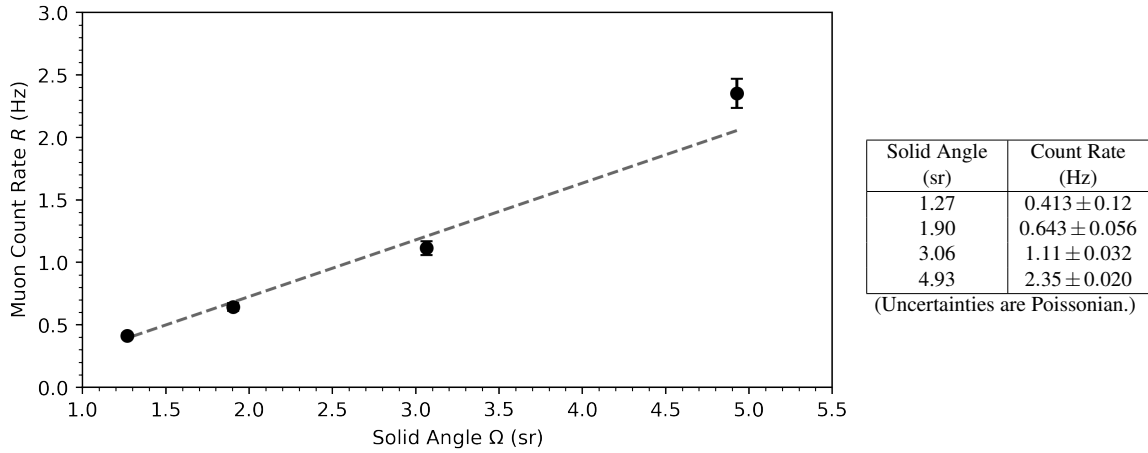


Figure 3: Data from the experiment.

On the 8th of October 2025, an experiment was conducted which followed the outlined theory in a physics lab at Maynooth University in isolation from any artificial sources of radiation.

When the muon count rate was plotted against the solid angle subtended by the scintillators, an approximately linear relation was found (see Figure 3). While the final data-point in the series is an outlier, the χ^2 -sum of the dataset is 0.049 (2 degrees of freedom), much less than the threshold value at a 95% level of significance (0.103), and so this point cannot be rejected from the data and rather must be accepted as a notable outlier.

The slope of this graph is equal to the **muon intensity** I_μ , which describes the rate at which muons flow through a unit solid angle.

$$I_\mu = 0.45 \pm 0.1 \text{ Hz sr}^{-1}$$

A good linear fit is expected from the data, as it implies a constant muon intensity and therefore an isotropic distribution, which was outlined in theory. However, further consideration introduces problems with the assumption that I_μ is a constant at all.

Further Consideration of the Model

At shallow angles of elevation, muons will have to travel through a larger distance of atmosphere, which increases the probability that they will decay or be absorbed. This implies that the actual data should fall beneath the linear regression for larger subtended solid angles. The experimental data seems to do the opposite, instead weakly implying that the rate might slightly increase at larger solid angles.

A more detailed statistical analysis also hints at underlying problems. If a second-order polynomial is fitted to the data, the chi-squared sum decreases by a factor of ten ($\chi^2 = 0.001$, 1 degree of freedom). Such a large difference in the chi-squared sum between polynomial orders and the almost perfect quadratic fit both point towards a problem with the model which, in some way, introduces an erroneous quadratic behaviour.

There is only one way in which theory affects the data displayed in Figure 3, and that's the way in which solid angle is calculated. Equation 1 is used to calculate this solid angle under the assumption that approximating the scintillators as circular introduces negligible error. However, it could be that a circle is not a good enough approximation of a rectangle at large solid angles. Only once an accurate solid angle is calculated can it be certain that trends in the data are "real" and not artefacts of the approximations used.

The equation for the accurate solid angle subtended by a rectangular plate of dimensions $a \times b$ at a distance d which has its normal vector pointed to the observer is possible to derive (Mathar, 2022)

$$\Omega(d) = 4 \arcsin \left(\frac{\alpha\beta}{\sqrt{(1+\alpha^2)(1+\beta^2)}} \right) : \alpha = \frac{a}{2d}, \beta = \frac{b}{2d} \quad (3)$$

which when manipulated will become

$$\Omega(d) = 4 \arcsin \left(\frac{ab}{\sqrt{(a^2 + 4d^2)(b^2 + 4d^2)}} \right) \quad (4)$$

When the solid angle data is then recalculated using Equation 4, the expected behaviour appears.

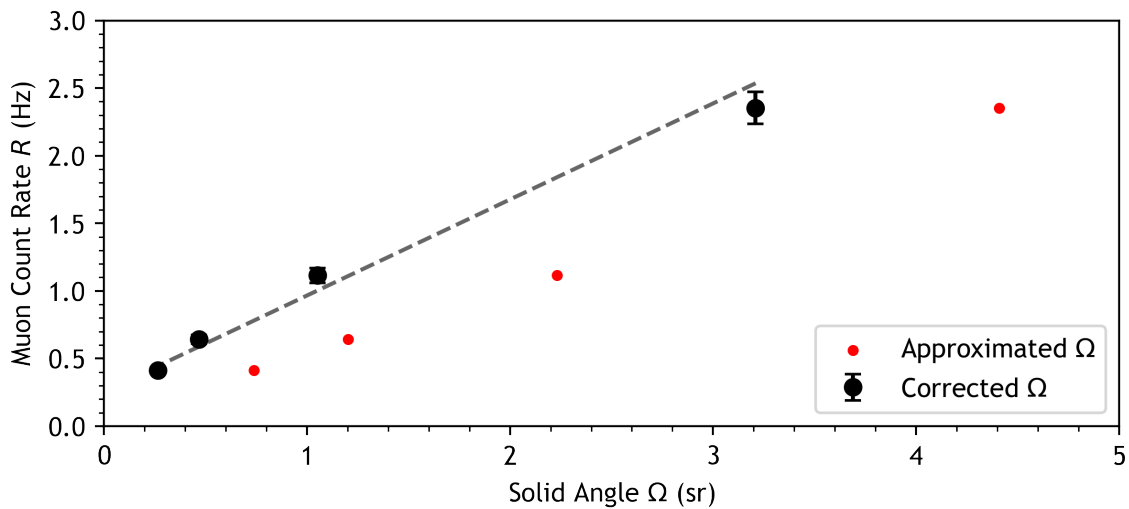


Figure 4: Experimental data with corrected solid angle calculation.

Analysis of Corrected Results

Once the solid angle has been corrected, a more informative analysis of the data displayed in Figure 4 becomes possible. As stated previously, a linear relationship between R and Ω is expected if the distribution of cosmic muons is isotropic. However, due to absorption and scattering in the atmosphere, fewer muons are expected to come from lower angles of elevation. It has been found that the rate of muons R_μ incident from a given zenithal angle θ is proportional to $\cos^2 \theta$ (Blanco *et al.*, 2008 and Blanchard, 2012).

Measuring the muons incident from a given solid angle is, in effect, taking the cumulative contribution of all angles from $\theta = 0$ to some $\theta = \theta_{\max}$, where $\theta_{\max}$ is the zenithal angle of an edge of the scintillator. The number of muons incident in a given solid angle should be proportional to the integral of $\cos^2 \theta$ across all angles which the solid angle subtends:

$$R \propto \int_0^{\theta_{\max}} \cos^2 \theta d\theta \implies R \propto \theta + \frac{\sin(2\theta)}{2}$$

The equation which relates the solid angle of a spherical cap to the zenithal angle is

$$\Omega = 2\pi(1 - \cos \theta)$$

As long as θ is not too small, this relationship is very close to linear ($\theta \approx A\Omega$). A spherical cap is actually the desired shape here, rather than rectangular, as technically there are two different zenithal angles to consider, that of the short edge and that of the long edge, so a spherical cap averages these angles out nicely.

$$\therefore R \propto A\Omega + \frac{\sin(2A\Omega)}{2} \quad (5)$$

The domain of this proportionality is $[0, \frac{\pi}{2})$, as beyond a zenithal angle of 90° , cosmic muons are blocked by the Earth itself.

When this proportionality is fitted to the data, a very close agreement is found, with $\chi^2 = 0.0016$ and one degree of freedom. In comparison, the chi-squared sum of a linear regression is more than ten times greater ($\chi^2 = 0.019$) albeit with an extra degree of freedom. Of course, when a fit agrees with experiment to such a great extent, there is always the risk that over-fitting has driven the chi-squared sum down. This is, in fact, an unavoidable factor to consider when the number of data-points is close to the number of parameters in the fitted function. This factor can be largely lessened by the measurement of more data-points, which time did not allow for in the original experiment.

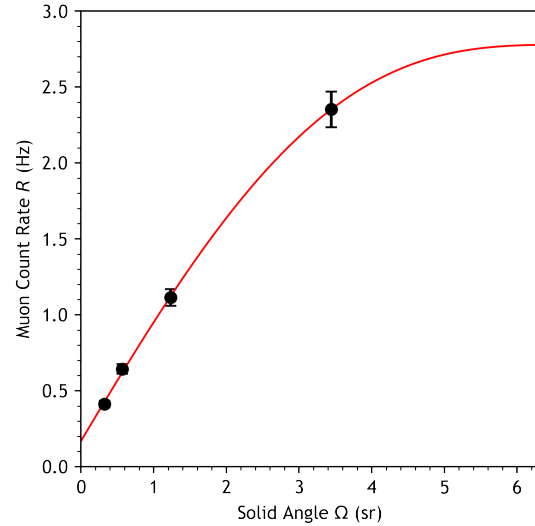


Figure 5: Fitting of Equation 5 to the data

The theoretical model fits the data very well, even compared to other possible fits, so it is likely that the relationship predicted in Equation 5 is accurate, with the caveat that more data is needed to confirm this, and remove doubt introduced by the risk of over-fitting.

Discussion

The results obtained in this experiment disagree slightly with the outlined theory, however once the attenuative affect of the atmosphere is accounted for, the results agree very well. The proportionality law $R \propto \cos^2 \theta$ has been found by some more recent studies to be a special case of a more general law; $R \propto \cos^{n-1} \theta$ where n is the power of the cosmic ray's energy spectrum (Shukla *et al.*, 2018). This means that the angular distribution of other secondary cosmic rays, such as protons and α -particles, will not necessarily be the same as that of muons. In fact, it now appears that the distribution of secondary cosmic protons is approximately $\cos^{1.93} \theta$ and for secondary cosmic α -particles is around $\cos^{1.75} \theta$ (Shukla *et al.*, 2018).

Due to the design of the detector in this experiment, it was impossible to probe the *azimuthal* angular distribution of cosmic muons. While initially, it seems as though the muon flux rate should have no dependence on the azimuthal angle, the interaction of the charged muons with the Earth's magnetic field has been found to have a significant effect on their distribution. In (Blanchard, 2012), a spark chamber was used in such a way that the azimuthal distribution of muons was revealed. In this study, the author found that there existed an asymmetry in the distribution of cosmic muons where there appeared to be a greater number of cosmic rays arriving from the west. The author took the deviation from simulation as a sign that something had gone wrong with the scintillators, however this is likely not the case, as the asymmetry has been noted in other similar independent studies. In particular, (Blanco *et al.* 2008) addressed the east-west asymmetry in their experiment and proposed the theory that the Earth's magnetic field was skewing the distribution. No matter what the reason may be, there is today substantial evidence indicating an underlying azimuthal dependence in the angular distribution of cosmic muons.

The number of data-points which could be taken was limited to the number of ways that the scintillators could be placed on the shelves, which was only four. In order to preform a more precise data analysis, more data-points are required, and so a different system to adjust the separation between the scintillators is needed. The measurement of smaller solid angles becomes increasingly difficult as there is a practical limit to how high the upper scintillator can be raised. Furthermore, there will exist a lower threshold of solid angle at which the coincidental detections which constitute noise in this system will dominate the actual signal. This implies that if a larger range of solid angles is probed, then a major change to the experimental set-up as a whole is needed, rather than the way in which the scintillators are separated alone.

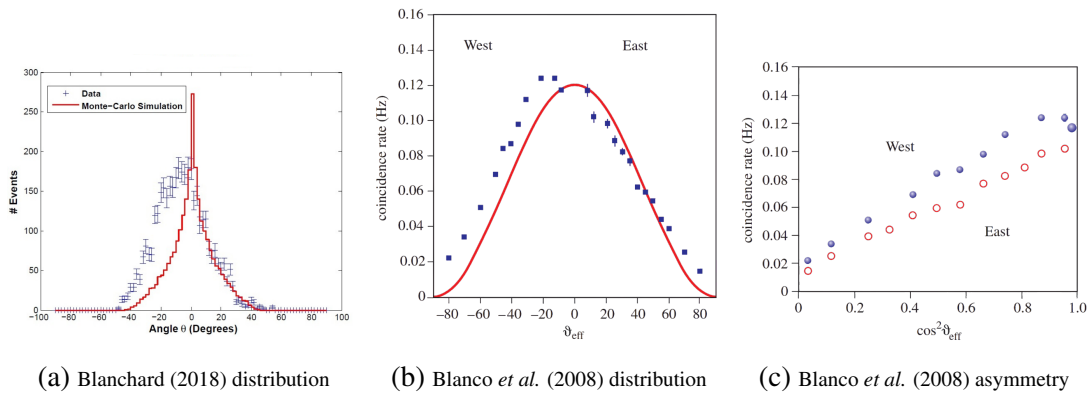


Figure 6: Examples of the azimuthal asymmetry in the distribution of charged cosmic rays

References

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Further calculations, including those of the χ^2 sum values used in this report can be found at <https://github.com/AQMU2024/Cosmic-Rays/blob/main/Cosmic%20Ray%20Detector.ipynb>